

Schneider Electric's digital twin, "Gemini," aims to create full virtual replicas of industrial plants. The initiative leverages AVEVA's software and Schneider's connected devices to create living digital models of physical assets.

for non-critical parameters, <5
3,200 edge computing nodes a
Kubernetes (K3s distribution) wi

KPI DASHBOARD — Schneider Electric Digital Twin Program																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
Metric										Target					Actual																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
										%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%					%				

3. Legrand SA — Smart Building IoT & Embedded Systems Strategy

Legrand SA, headquartered in Limoges, France, is a global specialist in electrical and digital building infrastructure with approximately 39,000 employees worldwide and annual revenue of €8.4 billion (FY2024). The company has been aggressively expanding its IoT capabilities, focusing on smart building systems that integrate lighting control, energy management, access control, and building automation into unified platforms. Legrand's "Eliot" (Electricity and IoT) program, launched in 2015 and now in its third phase, has connected over 40 million devices across commercial and residential installations globally.

Of particular significance for the technology labor market is Legrand's recently announced plan to hire 200 embedded engineers across its European R&D centers over the next 18 months. This hiring initiative, disclosed in a press release dated April 2025, focuses specifically on engineers with expertise in RTOS (Real-Time Operating Systems), BLE (Bluetooth Low Energy) mesh networking, Matter/Thread protocol stack implementation, and edge AI inference (specifically TinyML frameworks for resource-constrained microcontrollers). The positions span Legrand's R&D facilities in Limoges, Béziers, Strasbourg (France), Bergamo (Italy), and Arnhem (Netherlands). This represents one of the largest single embedded engineering hiring initiatives in the European electrical equipment sector and signals Legrand's commitment to internalizing IoT firmware development capabilities.

Legrand Connected Device Ecosystem

Product Line	Protocol	MCU Platform	Connectivity	Annual Vol.
Céliane with Netatmo	Zigbee 3.0	Nordic nRF52840	Wi-Fi bridge	2.1M units
Living Now (Bticino)	BLE Mesh	Silicon Labs EFR32	Cloud gateway	1.8M units
Eliot Pro	Matter/Thread	NXP i.MX RT	Native IP	450K units
Legrand ONE	MQTT/REST	Espressif ESP32-S3	Wi-Fi direct	800K units
IP PDU (Raritan)	SNMP/HTTPS	ARM Cortex-A7	Ethernet	120K units

Legrand's technology investment in Building Automation Systems (BAS) and Control Systems (BACS) D is driving growth in new commercial construction a and is expected to drive significant embedded engineering talent th 18% YoY in FY2024 to €412M in R&D development.

4. Dassault Systèmes SE — 3DEXPERIENCE & Virtual Twin Technology

Dassault Systèmes SE (DSY) employs 23,800 people and generated revenue of €15.1 billion in FY2024. Its 3DEXPERIENCE platform has evolved from a pure CAD platform to a multi-domain platform that spans design, simulation, and manufacturing between "digital twins" (static 3D models) and "virtual twins" (dynamic, physics-based real-time).

The 3DEXPERIENCE platform
both AWS and Outscale, Dassault
GPU-accelerated compute clusters
Apriso" for manufacturing operations
connection between factory floor
components include: Abaqus finite
method (fluid dynamics), CST
based fluid simulation).

VIRTUAL TWIN DEPLOYMENT METRICS – Dassault Systèmes	
Deployment Model	Statistics
Cloud (3DEXP on AWS/Outscale)	4,200 enterprise tenants
On-premise (private cloud)	1,100 installations
Hybrid (edge + cloud)	680 deployments
Total connected IoT sensors	~8.2M (via OPC-UA/MQTT)
Simulation jobs / month	12.4M
GPU hours consumed / month	47M (A100/H100 equiv.)
Active virtual twin models	23,400

Platform Comparison: Industrial Digital Twin Solutions

Vendor	Platform	Strengths	Protocols	Pricing Model
Dassault Systèmes	3DEXPERIENCE	Multi-physics sim, PLM	OPC-UA, REST	Per-user + compute
Siemens	Xcelerator/MindSphere	Factory automation, MES	OPC-UA, MQTT	Subscription + usage
PTC	ThingWorx + Creo	AR integration, CAD	MQTT, REST	Subscription
Schneider (AVEVA)	PI System + E3D	Process industries	OPC-UA, PI AF	Per-server license
ABB	ABB Ability	Drives & robotics	MQTT, OPC-UA	Per-device + cloud

5. Edge Computing Infrastructure Analysis

The deployment of edge computing is reaching an inflection point, with our survey indicating that 65% of manufacturers have deployed at least one edge device: (1) field-level gateway (e.g., Siemens Profinet I RT/MQTT/OPC-UA), (2) K3s or MicroK8s on x86 hardware for local facilities for fleet-level analytics (e.g., ABB), while Legrand's IoT solutions provide a cloud-based Predictive maintenance representation. 70% of manufacturers reporting average of 10 edge deployments. The technology stack includes sensors connected via MQTT, AI models, with model training on the edge. EcoStruxure Asset Performance Monitor tools represent complementary offerings. Dassault's approach leverages predictive analytics.